

Homestead Unit Plans & Academic Integration Guide

A Companion Curricular Extension of the Homestead Academic Mapping Framework

This guide outlines step-by-step modular units designed to satisfy Missouri high school transcript standards under RSMo § 167.012 through hands-on mastery. By operationalizing our core ecosystem principle—**Real-Life Work yields Multi-Subject Academic Outcomes**—parents can seamlessly translate everyday homesteading tasks into formal core and elective high school credits.

Unit 1: The Real-World Pantry Audit (Math & Economics)

Target Grade: 5th-8th Grade or High School (Applied Mathematics & Practical Life Skills)
Objective: Students learn budgeting, percentage calculations, unit rate comparison, and resource management through a structured, multi-step household pantry inventory audit.

- **Step 1: Ingest & Inventory** — Count and categorize household food items by type, shelf life, and packaging weight.
- **Step 2: Unit Cost Calculation** — Identify the total purchase price of five key bulk items and compute the price per ounce/gram (unit rate).
- **Step 3: Data Representation** — Construct a visual chart or graph showing food category proportions (e.g., canned goods, dry goods, fresh produce).
- **Step 4: Waste & Savings Strategy** — Identify one frequently wasted grocery item and calculate the precise yearly cost savings of replacing it with an organic, home-preserved homestead alternative.

Unit 2: Rainwater Catchment & Filtration Test (Science & Lab)

Target Grade: High School / Adult Learning (Homesteading & Off-Grid Physics Laboratory)
Objective: Students design, build, and analyze a low-cost rainwater catchment and multi-stage biological filtration mechanism to study fluid dynamics, microbiology, and water chemistry.

Parameter / Step	Measurement / Observation	Notes & Scientific Analysis
Setup Cost	Under \$20 Total	Constructed using a recycled bucket, gravel, coarse sand, and activated charcoal.
Pre-Filter Clarity	Turbid (Sediment visible)	Collected after initial roof run-off; organic particles suspended.
Post-Filter Clarity	Clear & Sediment-Free	Particle filtration successful; multi-layered mechanical filtration completed.
Practical Application	Non-potable home use	Ideal for organic garden irrigation and non-drinking livestock watering uses.

Audit-Proof Documentation Framework

Under Missouri state guidelines, keeping clean portfolios of work samples is critical. When your student completes these unit studies, you should file the following items in their high school portfolio binder:

- **Completed worksheets and formulas** showing unit cost conversions and pantry savings calculations.
- **A physical diagram** detailing the layers of the rainwater filtration bucket and a written laboratory report.
- **Dated, chronological log hours** entered on the *Missouri Homestead Hours Log Sheet* to confirm home-based core compliance.

Legal Notice: Pursuant to RSMo § 167.012, homeschooling in Missouri requires 1,000 hours of annual instruction, with 600 hours in core areas (such as math and science) and 400 of those core hours occurring at home. By integrating these unit studies with the *Missouri Hours Log Spreadsheet*, you satisfy all requirements while providing your student with invaluable real-world experience.